

OCP Java SE 21 Developer — Chapter 6 Practice Quiz

Class Design | 8 Questions | Multiple Choice

Name: _____ Date: _____

1. Given the following code, what is the result?

```
public class Animal {  
    public Animal(int age) { System.out.print("A"); }  
}  
public class Dog extends Animal {  
    public Dog() { System.out.print("D"); }  
}  
public class Main {  
    public static void main(String[] args) { new Dog(); }  
}
```

- ☐ A. AD
- ☐ B. DA
- ☐ C. D
- ☐ D. The code does not compile.

2. What is printed when main() is executed?

```
public class Tank {  
    static { System.out.print("S"); }  
    { System.out.print("I"); }  
    public Tank() { System.out.print("C"); }  
    public static void main(String[] args) {  
        new Tank();  
        new Tank();  
    }  
}
```

- ☐ A. SICSIC
- ☐ B. SICC
- ☐ C. SICIC
- ☐ D. The code does not compile.

3. Which of the following statements are true about abstract classes and abstract methods? (Choose all that apply.)

- ☐ A. An abstract class must declare at least one abstract method.
- ☐ B. An abstract class can be instantiated if it defines a public constructor.
- ☐ C. A non-abstract class that extends an abstract class must implement all inherited abstract methods.
- ☐ D. A method cannot be marked both abstract and final.
- ☐ E. An abstract method may include a body enclosed in braces.

4. Which of the following are valid overrides of display() in Child? (Choose all that apply.)

```
class Parent {  
    protected void display() {}  
}  
class Child extends Parent {  
    // insert override here  
}
```

- ☐ A. public void display() {}
- ☐ B. protected void display() {}
- ☐ C. private void display() {}
- ☐ D. void display() {} // package-private

5. Which of the following statements about `this()` and `super()` in constructors are correct? (Choose all that apply.)

- ☐ A. Both `this()` and `super()` can appear in the same constructor.
- ☐ B. `this()` must be the first statement in a constructor if used.
- ☐ C. If a constructor contains neither `this()` nor `super()`, the compiler inserts `super()` automatically.
- ☐ D. `super()` always refers to the constructor of `java.lang.Object`.
- ☐ E. A constructor may call `this()` and `super()` in any order.

6. Consider the following code. Which lines cause a compilation error? (Choose all that apply.)

```
public class Bird {  
    public static void fly() {} // line 1  
    public void sing() {} // line 2  
}  
public class Parrot extends Bird {  
    public void fly() {} // line 3  
    public static void sing() {} // line 4  
}
```

- ☐ A. Line 1
- ☐ B. Line 2
- ☐ C. Line 3
- ☐ D. Line 4

7. Which declarations in `Parrot` are valid overrides of `getName()` in `Bird`? (Choose all that apply.)

```
public class Bird {  
    protected CharSequence getName() { return "Bird"; }  
}  
public class Parrot extends Bird {  
    // insert override here  
}
```

- ☐ A. `public String getName() { return "Parrot"; }`
- ☐ B. `public CharSequence getName() { return "Parrot"; }`
- ☐ C. `public Object getName() { return "Parrot"; }`
- ☐ D. `protected StringBuilder getName() { return new StringBuilder("Parrot"); }`

8. Given the following code, what is the result?

```
public class Vehicle {  
    private final int speed;  
    private final String type;  
    { speed = 100; }  
    public Vehicle(String type) { this.type = type; }  
    public Vehicle() { // line A  
        this.speed = 50; // line B  
    }  
}
```

- ☐ A. The code compiles successfully.
- ☐ B. Does not compile: `speed` is assigned twice in the second constructor.
- ☐ C. Does not compile: `type` is never assigned in the second constructor.
- ☐ D. Does not compile: final fields cannot be assigned in instance initializers.